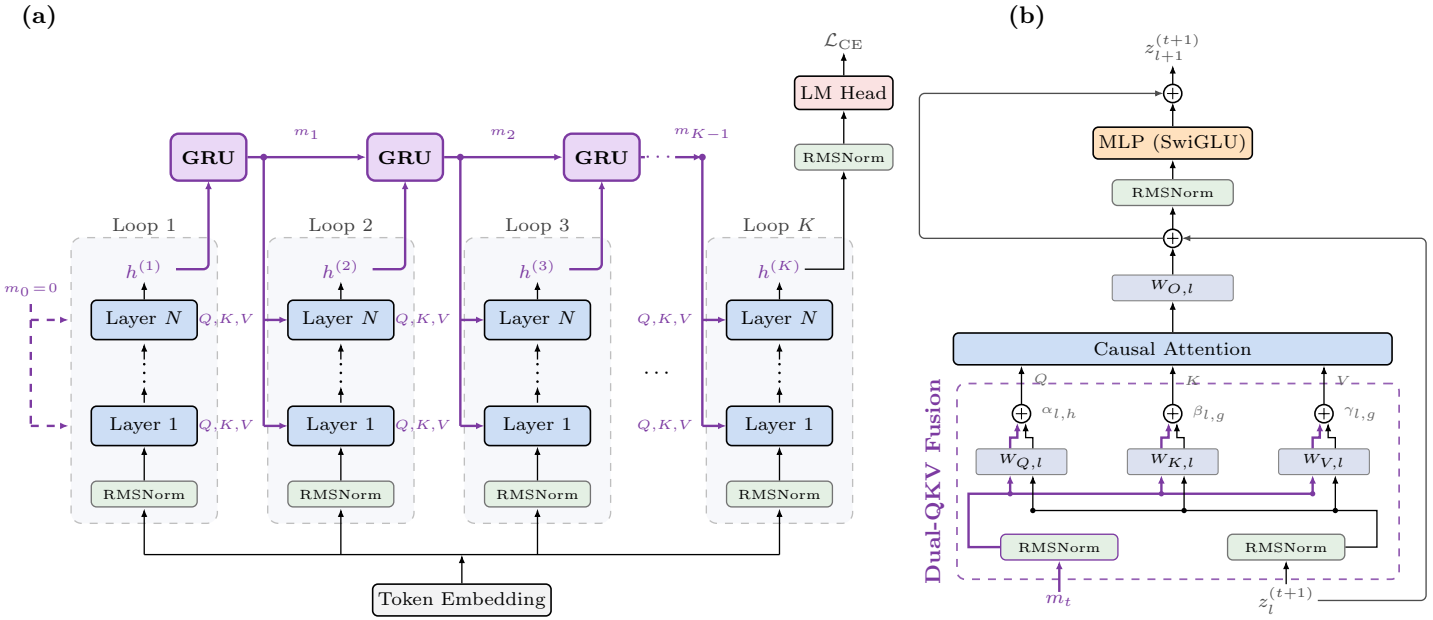

(a)
$\mathcal{L}_{\text{CE}}$
LM Head
RMSNorm
GRU
$m_1$
GRU
$m_2$
GRU
$m_{K-1}$
Loop 1
Loop 2
Loop 3
Loop $K$
$h^{(1)}$
$h^{(2)}$
$h^{(3)}$
$h^{(K)}$
$m_0 = 0$
Layer $N$
Layer 1
$Q, K, V$
RMSNorm
Token Embedding
(b)
$z_{l+1}^{(t+1)}$
MLP (SwiGLU)
RMSNorm
$W_{O,l}$
Causal Attention
$Q$
$K$
$V$
$\alpha_{l,h}$
$\beta_{l,g}$
$\gamma_{l,g}$
$W_{Q,l}$
$W_{K,l}$
$W_{V,l}$
Dual-QKV Fusion
RMSNorm
RMSNorm
$m_t$
$z_l^{(t+1)}$